

Airport Lost Luggage Claim & Tracking Portal

Lab 1 - Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass: A valid lost-baggage report with a matching scan record identifies the bag and generates a recovery alert. Fail: The system accepts an incorrect bag-tag format.	Enables accurate identification and recovery of lost baggage.
FR-002	Functional	The system shall allow a passenger to submit a lost-baggage claim using a valid bag tag, flight details, and contact information.	High	Pass: When all required fields contain valid values, the system creates a claim and displays a claim reference number. Fail: The system creates a claim when a required field is missing or the bag tag is invalid.	Provides the information needed to begin baggage recovery.
FR-003	Functional	The system shall allow a passenger to view the current status and available scan timeline of a submitted lost-baggage claim.	High	Pass: A passenger who provides a valid claim reference can view the current claim status and recorded scan events. Fail: The system displays another passenger's claim details or an incorrect claim status.	Keeps passengers informed during the recovery process.
FR-004	Functional	The system shall allow a baggage service agent to review a claim, record the baggage recovery status, and update the claim status.	High	Pass: An authorized agent can update a valid claim to a recovery status, and the revised status is visible when the claim is viewed. Fail: An unauthorized user can update a claim, or the update is not saved.	Supports operational handling of baggage recovery.
FR-005	Functional	The system shall automatically initiate the passenger compensation workflow when a claim meets the airline's configured compensation conditions.	Medium	Pass: For a claim that meets the configured conditions, the system records that compensation processing has started. Fail: The system initiates compensation for a claim that does not meet the conditions.	Supports timely and consistent compensation handling.
NFR-001	Non-Functional (Performance & Security)	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load. Fail: The complete scan history is not returned within 200 ms under the test conditions.	Ensures timely, secure access to baggage tracking information.
NFR-002	Non-Functional (Security)	The system shall restrict claim details and claim-status updates to authenticated passengers for their own claims and authorized baggage service agents.	High	Pass: Authorized users can access only permitted claim data and functions; unauthorized access or update attempts are denied. Fail: A passenger can access another passenger's claim, or an unauthenticated user can update a claim.	Protects passenger information and prevents unauthorized claim changes.